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## INFORMATION PROCESSING APPARATUS, RADIOGRAPHY SYSTEM, INFORMATION PROCESSING METHOD, AND PROGRAM


### Abstract

Provided are an information processing apparatus, a radiography system, an information processing method, and a program capable of supporting determination of whether or not a specific part of a subject is included in an irradiation range of radiation. The information processing apparatus includes at least one processor. The processor is configured to: recognize a specific part of a subject from an optical image obtained by capturing the subject; derive a distance to the recognized specific part using a distance image obtained by capturing the subject; and derive an irradiation range of radiation at a position corresponding to the derived distance in an irradiation axis direction.



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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. 2024-025788, filed on Feb. 22, 2024. The above application is hereby expressly incorporated by reference, in its entirety, into the present application.

BACKGROUND

1. Technical Field

[0002] The disclosed technology relates to an information processing apparatus, a radiography system, an information processing method, and a program.

2. Description of the Related Art

[0003] Regarding a technology of controlling an irradiation range of radiation in a radiography apparatus, the following technology is known. For example, in description of JP2021-191403A, whether or not a structure of a specific shape having a transmittance of radiation lower than that of a subject is present in an imaging region of a radiography apparatus is specified based on the specific shape, and in a case where the structure is present, control is performed such that a region excluding the structure is the imaging region.

[0004] JP2017-534401A describes a pre-X-ray exposure control device including a subject detection unit, a subject model unit, an interface unit, a processing unit, and a display unit. The subject detection unit detects subject data of a subject to be exposed. The subject model unit provides a subject model and refines the subject model based on the subject data to obtain a refined subject model. The interface unit provides setting data of an X-ray unit to be used for exposing the subject. The processing unit calculates a virtual X-ray projection image based on the refined subject model and the provided setting data. The display unit displays the virtual X-ray projection image. The setting data is a collimation parameter of an X-ray unit to be used for exposing a partial region of the subject. The subject detection unit detects a position of an anatomical landmark of the subject and detects an orientation of the subject based on the position of the anatomical landmark.

SUMMARY

[0005] In a case where a radiation image is captured for a subject, an irradiation range of radiation may be adjusted such that a specific part of the subject is not irradiated with the radiation. For example, in a videofluoroscopic examination of swallowing using a radiography apparatus, an irradiation range of radiation is adjusted such that an eyeball of the subject is not irradiated with the radiation.

[0006] In the adjustment of the irradiation range of the radiation, for example, the irradiation range of the radiation is confirmed by irradiating the subject with collimated light for displaying the irradiation range of the radiation. However, in a case where an uneven part (for example, nose and mouth) of the subject is included in the irradiation range of the collimated light, an edge position of the collimated light is unclear. As a result, it is difficult to determine whether or not a specific part of the subject, which is to be avoided from the irradiation with the radiation, is included in the irradiation range of the radiation.

[0007] The disclosed technology has been made in view of the above-described points, and an object of the disclosed technology is to support a determination of whether or not the specific part of the subject is included in the irradiation range of the radiation.

[0008] According to the disclosed technology, there is provided an information processing apparatus comprising at least one processor. The processor is configured to: recognize a specific part of a subject from an optical image obtained by capturing the subject; derive a distance to the recognized specific part using a distance image obtained by capturing the subject; and derive an irradiation range of radiation at a position corresponding to the derived distance in an irradiation axis direction.
[0009] The processor may be configured to display information indicating the derived irradiation range on a display screen, by superimposing the information on the optical image. The processor may be configured to display the optical image and a radiation image side by side. The processor may be configured to issue an alert in a case where the recognized specific part is included in the derived irradiation range.
[0010] The processor may be configured to control the irradiation range of the radiation such that the recognized specific part is not included in the irradiation range of the radiation in accordance with a command, in a case where the recognized specific part is included in the derived irradiation range. The processor may be configured to control the irradiation range of the radiation by controlling an opening amount of an opening portion, through which the radiation passes, of a collimator that forms the opening portion. The specific part may be an eyeball of the subject.
[0011] A radiography system according to the disclosed technology includes the above-described information processing apparatus, an optical camera that generates an optical image, a distance sensor that generates a distance image, and a radiation source unit that emits radiation.
[0012] The optical camera may be disposed outside an irradiation range of the radiation emitted from the radiation source unit. The optical camera may be disposed at a position closer to an irradiation axis of the radiation emitted from the radiation source unit with respect to the distance sensor. The distance sensor may be a ToF camera or a stereo camera.
[0013] An information processing method according to the disclosed technology is a method executed by at least one processor included in an information processing apparatus, the method including a process comprising: recognizing a specific part of a subject from an optical image obtained by capturing the subject; deriving a distance to the recognized specific part using a distance image obtained by capturing the subject; and deriving an irradiation range of radiation at a position corresponding to the derived distance in an irradiation axis direction.
[0014] A program according to the disclosed technology is a program that causes at least one processor included in an information processing apparatus, to execute a process comprising: recognizing a specific part of a subject from an optical image obtained by capturing the subject; deriving a distance to the recognized specific part using a distance image obtained by capturing the subject; and deriving an irradiation range of radiation at a position corresponding to the derived distance in an irradiation axis direction.
[0015] According to the disclosed technology, it is possible to support the determination of whether or not the specific part of the subject is included in the irradiation range of the radiation.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0016] Exemplary embodiments according to the technique of the present disclosure will be described in detail based on the following figures, wherein:
[0017] FIG. **1** is a diagram showing an example of a configuration of a radiography system according to an embodiment of disclosed technology;
[0018] FIG. **2** is a view showing an example of a configuration of a radiography apparatus according to the embodiment of the disclosed technology;
[0019] FIG. **3** is a diagram showing an example of a relationship between an irradiation range of

radiation and an imaging range of an optical camera according to the embodiment of the disclosed technology;
[0020] FIG. **4**A is a diagram showing a positional relationship between the optical camera and a distance sensor according to the embodiment of the disclosed technology;
[0021] FIG. **4**B is a diagram showing the positional relationship between the optical camera and the distance sensor according to the embodiment of the disclosed technology;
[0022] FIG. **5** is a diagram showing an example of a hardware configuration of an information processing apparatus according to the embodiment of the disclosed technology;
[0023] FIG. **6** is a functional block diagram showing an example of a functional configuration of the information processing apparatus according to the embodiment of the disclosed technology;
[0024] FIG. **7** is a view for describing a method of deriving an irradiation range of radiation according to the embodiment of the disclosed technology;
[0025] FIG. **8** is a diagram showing an example of a display form of information indicating the irradiation range of the radiation according to the embodiment of the disclosed technology;
[0026] FIG. **9** is a diagram showing an example of a relationship between an irradiation range of radiation and an imaging range of an optical camera;
[0027] FIG. **10** is a flowchart showing an example of a flow of processing performed by executing a processing program by a CPU of the information processing apparatus according to the embodiment of the disclosed technology;
[0028] FIG. **11** is a functional block diagram showing an example of a functional configuration of an information processing apparatus according to another embodiment of the disclosed technology;
[0029] FIG. **12** is a flowchart showing an example of a flow of processing performed by executing a processing program by a CPU of the information processing apparatus according to another embodiment of the disclosed technology;
[0030] FIG. **13** is a functional block diagram showing an example of the functional configuration of the information processing apparatus according to another embodiment of the disclosed technology;
[0031] FIG. **14** is a perspective view showing an example of a configuration of a collimator according to the embodiment of the disclosed technology; and
[0032] FIG. **15** is a flowchart showing an example of a flow of processing performed by executing a processing program according to another embodiment of the disclosed technology.

## DETAILED DESCRIPTION

[0033] An example of embodiments of the disclosed technology will be described below with reference to the drawings. It is noted that the same or equivalent components and portions in the drawings are assigned by the same reference numerals, and the overlapping description will be omitted.

### First Embodiment

[0034] FIG. **1** is a diagram showing an example of a configuration of a radiography system **1** according to an embodiment of the disclosed technology. The radiography system **1** includes an information processing apparatus **10**, a radiography apparatus **20**, an optical camera **30**, and a distance sensor **40**.

[0035] FIG. **2** is a diagram showing an example of the configuration of the radiography apparatus **20**. The radiography apparatus **20** is capable of imaging a radiation image using radiation such as X-rays. The radiography apparatus **20** has an imaging table **21**. The imaging table **21** is supported on the floor surface by a stand **22**. A radiation source unit **24** is attached to the imaging table **21** via a support column **23**. The radiation source unit **24** is configured to include a radiation source **25** and a collimator **26**. The imaging table **21** has a built-in radiation detector **27**.

[0036] The radiation source **25** has a radiation tube (not shown) that emits radiation. The collimator **26** limits an irradiation range of the radiation emitted from the radiation tube. The radiation detector **27** has a plurality of pixels that generate signal charges in accordance with the radiation that has been transmitted through a subject P. The radiation detector **27** is called a flat panel detector (FPD).

[0037] The radiation source unit **24** is capable of reciprocating together with the support column **23** along a long side direction of the imaging table **21** by a moving mechanism (not shown) such as a motor. The radiation detector **27** is capable of reciprocating along the long side direction of the imaging table **21** in conjunction with the movement of the radiation source unit **24**. The imaging table **21** and the support column **23** can be rotated between a standing state shown in FIG. **2** and a supine state (not shown) by a rotation mechanism (not shown) such as a motor. In the supine state, the surface of the imaging table **21** is parallel to the floor surface, and the direction in which the support column **23** extends is perpendicular to the floor surface. On the other hand, in the standing state, the surface of the imaging table **21** is perpendicular to the floor surface, and the direction in which the support column **23** extends is parallel to the floor surface. In the standing state, it is possible to perform fluoroscopy on the subject P in a wheelchair **50** as shown in FIG. **2**. In the following description, as shown in FIG. **2**, a case (LR imaging) where the radiation is emitted from a side of the subject P to capture the radiation image will be described as an example. In addition, an irradiation axis direction of the radiation is referred to as a Y direction, a vertical direction is referred to as a Z direction, and a front-back direction of the subject P is referred to as an X direction.

[0038] The optical camera **30** is an imaging apparatus that can generate an optical image (RGB image). An optical image of the subject P is captured by the optical camera **30**. The optical camera **30** is attached to the radiation source unit **24**. In FIG. **2**, a case where the optical camera **30** is attached to a housing of the collimator **26** is shown, but the optical camera **30** may be attached to a housing of the radiation source **25**.

[0039] FIG. **3** is a diagram showing an example of a relationship between an irradiation range Ra of the radiation emitted from a radiation tube **28** and an imaging range Rb of the optical camera **30**. In order to avoid the optical camera **30** from being captured on the radiation image, the optical camera **30** is disposed outside the irradiation range Ra of the radiation. Therefore, an optical axis of the optical camera **30** and an irradiation axis of the radiation are parallel to each other but are not matched. The entire irradiation range Ra of the radiation at a position of the subject P is included in the imaging range Rb of the optical camera **30**.

[0040] The distance sensor **40** is a device that can generate a distance image showing a distance to a surface of an object. A distance image of the subject P is captured by the distance sensor **40**. The distance sensor **40** may be, for example, a time-of-flight (ToF) camera that generates a distance image with a ToF method using infrared light. In addition, the distance sensor **40** may be a stereo camera that generates a distance image based on a principle of triangulation using two cameras. The distance sensor **40** is attached to the radiation source unit **24**, similarly to the optical camera **30**. In FIG. **2**, a case where the distance sensor **40** is attached to the housing of the collimator **26** is shown, but the distance sensor **40** may be attached to the housing of the radiation source **25**.

[0041] FIGS. **4**A and **4**B are diagrams showing a positional relationship between the optical camera **30** and the distance sensor **40**, respectively. FIG. **4**A shows a case where the distance sensor **40** is a ToF camera, and FIG. **4**B shows a case where the distance sensor **40** is a stereo camera. The optical camera **30** is disposed at a position closer to an irradiation axis AX of the radiation emitted from the radiation tube **28** with respect to the distance sensor. In a case where the distance sensor **40** is a stereo camera having two cameras **41** and **42**, as shown in FIG. **4**B, the optical camera **30** may be disposed between the two cameras **41** and **42** constituting the distance sensor **40**. It is preferable that a position of the optical axis of the optical camera **30** and a position of the irradiation axis AX of the radiation match with each other in the Z direction (vertical direction).

[0042] The information processing apparatus **10** has a function of deriving an irradiation range of radiation at a position on an irradiation axis where a specific part of the subject P is present, based on the optical image of the subject P output from the optical camera **30** and the distance image of the subject P output from the distance sensor **40**. The “specific part” is a predetermined part as a part to be avoided from irradiation with radiation among parts of body of the subject P. In the following, a case where the specific part to be avoided from the irradiation with the radiation is an eyeball will be described as an example.

[0043] FIG. **5** is a diagram showing an example of a hardware configuration of the information processing apparatus **10**. The information processing apparatus **10** includes a central processing unit (CPU) **101**, a random-access memory (RAM) **102**, a non-volatile memory **103**, an input device **104** including a keyboard and a mouse, a display **105**, and a communication interface **106**. These pieces of hardware are connected to a bus **108**.

[0044] The display **105** may be a touch panel display. The communication interface **106** is an interface for the information processing apparatus **10** to perform communication with the optical camera **30**, the distance sensor **40**, and the radiography apparatus **20**. The communication method may be either wired or wireless. For the wireless communication, it is possible to apply a method conforming to existing wireless communication standards such as, for example, Wi-Fi (registered trademark) and Bluetooth (registered trademark).

[0045] The non-volatile memory **103** is a non-volatile storage medium such as a hard disk or a flash memory. A processing program **110** is stored in the non-volatile memory **103**. The RAM **102** is a work memory for the CPU **101** to execute processing. The CPU **101** loads the processing program **110** stored in the non-volatile memory **103** into the RAM **102**, and executes processing in accordance with the processing program **110**. The CPU **101** is an example of a “processor” in the disclosed technology.

[0046] FIG. **6** is a functional block diagram showing an example of a functional configuration of the information processing apparatus **10**. The information processing apparatus **10** includes an optical image acquisition unit **11**, a distance image acquisition unit **12**, a specific part recognition unit **13**, a distance derivation unit **14**, an irradiation range derivation unit **15**, and a display processing unit **16**. By executing the processing program **110** by the CPU **101**, the information processing apparatus **10** functions as the optical image acquisition unit **11**, the distance image acquisition unit **12**, the specific part recognition unit **13**, the distance derivation unit **14**, the irradiation range derivation unit **15**, and the display processing unit **16**.

[0047] The optical image acquisition unit **11** acquires the optical image of the subject P output from the optical camera **30**. The distance image acquisition unit **12** acquires the distance image of the subject P output from the distance sensor **40**.

[0048] The specific part recognition unit **13** recognizes the eyeball, which is the specific part to be avoided from the irradiation with the radiation, among the parts of the body of the subject P from the optical image acquired by the optical image acquisition unit **11**. The specific part recognition unit **13** recognizes the specific part (eyeball) from the optical image using a discriminator constructed by machine learning using the optical image, in which correct answer information is attached to the specific part, as training data.

[0049] The distance derivation unit **14** derives a distance to the specific part (eyeball) recognized by the specific part recognition unit **13** by using the distance image acquired by the distance image acquisition unit **12**. More specifically, the distance derivation unit **14** transforms a coordinate position on the optical image where the specific part (eyeball) recognized by the specific part recognition unit **13** is present, into a coordinate position on the distance image. This coordinate transformation can be performed by geometric calculation based on installation positions of the optical camera **30** and the distance sensor **40**. The distance derivation unit **14** derives the distance to the specific part (eyeball) from a pixel value at the coordinate position on the distance image in which the specific part (eyeball) is present.

[0050] The irradiation range derivation unit **15** derives an irradiation range of radiation at a position corresponding to the distance to the specific part (eyeball) derived by the distance derivation unit **14**, in the irradiation axis direction (Y direction). FIG. **7** is a diagram for describing a method of deriving an irradiation range of radiation. The radiation emitted from the radiation tube **28** is limited to a range of an opening portion **60** of the collimator disposed in front of the radiation tube **28**. Dimensions a and b of the opening portion **60** and a distance c from the radiation tube **28** to the opening portion **60** are known. A distance D from the radiation tube **28** to the specific part (eyeball) of the subject P can be acquired by performing correction in accordance with a positional relationship between the distance sensor **40** and the radiation tube **28**, with respect to the distance to the eyeball derived by the distance derivation unit **14**. The irradiation range derivation unit **15** derives the irradiation range of the radiation at the position corresponding to the distance to the eyeball, in the irradiation axis direction (Y direction), by calculating W represented by the following Expression (1) and T represented by the following Expression (2). That is, the irradiation range derivation unit **15** derives a rectangular region having a length of W in the X direction and a length of T in the Z direction with the irradiation axis AX of the radiation as a center, as the irradiation range of the radiation at the eyeball position of the subject P.

[00001] $W = a \times D / c \quad (1) \quad T = b \times D / c \quad (2)$

[0051] The display processing unit **16** displays information indicating the irradiation range of the radiation derived by the irradiation range derivation unit **15** on a display screen of the display **105**, by superimposing the information on the optical image of the subject P acquired by the optical image acquisition unit **11**. FIG. **8** is a diagram showing an example of a display form of information indicating an irradiation range of radiation displayed on a display screen **200**. The display screen **200** may be a screen of a console installed outside an imaging room or may be a display screen of a monitor cart provided in the imaging room.

[0052] The display processing unit **16** displays a bounding box **211** as information indicating the irradiation range of the radiation (that is, the irradiation range of the radiation at the position corresponding to the distance to the eyeball) derived by the irradiation range derivation unit **15**, in the irradiation axis direction (Y direction), on an optical image **210** of the subject P acquired by the optical image acquisition unit **11**, by superimposing the information. As shown in FIG. **8**, the display processing unit **16** may display a radiation image **220** and the optical image **210** of the subject P side by side. The radiation image **220** may be a fluoroscopic image (video) acquired by fluoroscopy using the radiography apparatus **20**. The optical image **210** and the radiation image **220** are displayed on the display screen **200** in real time.

[0053] FIG. **9** is a diagram showing an example of a relationship between an irradiation range of the radiation emitted from the radiation tube **28** and an imaging range of the optical camera **30**. As described above, since the optical axis of the optical camera **30** and the irradiation axis of the radiation are not matched, a position and a size of the irradiation range of the radiation within the imaging range of the optical camera **30** change in accordance with a distance from the radiation tube **28**. That is, a position and a size of an irradiation range R.sub.a1 of radiation at a position at a distance D.sub.1 from the radiation tube **28** within an imaging range R.sub.b1 of the optical camera **30** at the same position are different from a position and a size of an irradiation range R.sub.a2 of radiation at a position at a distance D.sub.2 from the radiation tube **28** within an imaging range R.sub.b2 of the optical camera **30** at the same position.

[0054] In consideration of the above-described points, in a case where the bounding box **211** is displayed on the optical image **210** in a superimposed manner, the display processing unit **16** derives a position and a size of the bounding box **211** on the optical image **210**, based on the distance (distance D from the radiation tube **28** to the eyeball of the subject P) derived by the distance derivation unit **14**. The position and size of the bounding box **211** on the optical image **210** can be derived by performing geometric calculation in consideration of a positional relationship between the radiation tube **28** and the optical camera **30**.

[0055] A capturer of the radiation image can ascertain whether or not the specific part (eyeball) of the subject P is included in the irradiation range of the radiation by confirming the display screen

**200** shown in FIG. **8** as an example.

[0056] FIG. **10** is a flowchart showing an example of a flow of processing performed by the CPU **101** executing the processing program **110**.

[0057] In step S1, the optical image acquisition unit **11** acquires the optical image of the subject P output from the optical camera **30**.

[0058] In step S2, the distance image acquisition unit **12** acquires the distance image of the subject P output from the distance sensor **40**.

[0059] In step S3, the specific part recognition unit **13** recognizes the eyeball, which is the specific part to be avoided from the irradiation with the radiation, among the parts of the body of the subject P from the optical image acquired in step S1.

[0060] In step S4, the distance derivation unit **14** derives the distance to the specific part (eyeball) recognized in step S3 by using the distance image acquired in step S2.

[0061] In step S5, the irradiation range derivation unit **15** derives an irradiation range of radiation at a position corresponding to the distance to the specific part (eyeball) derived in step S4, in the irradiation axis direction (Y direction).

[0062] In step S6, the display processing unit **16** displays the information indicating the irradiation range of the radiation derived in step S5, on the display screen of the display **105** by superimposing the information on the optical image of the subject P acquired in step S1.

[0063] As described above, the information processing apparatus **10** according to the embodiment of the disclosed technology recognizes a specific part of a subject from an optical image obtained by capturing the subject, derives a distance to the recognized specific part using a distance image obtained by capturing the subject, derives an irradiation range of radiation at a position corresponding to the derived distance in an irradiation axis direction, and displays information indicating the derived irradiation range on a display screen by superimposing the information manner on the optical image.

[0064] In a case where a radiation image is captured for a subject, an irradiation range of radiation may be adjusted such that a specific part of the subject is not irradiated with the radiation. For example, in a videofluoroscopic examination of swallowing using a radiography apparatus, an irradiation range of radiation is adjusted such that an eyeball of the subject is not irradiated with the radiation.

[0065] In the adjustment of the irradiation range of the radiation, for example, the irradiation range of the radiation is confirmed by irradiating the subject with collimated light for displaying the irradiation range of the radiation. However, in a case where an uneven part (for example, nose and mouth) of the subject is included in the irradiation range of the collimated light, an edge position of the collimated light is unclear. As a result, it is difficult to determine whether or not a specific part of the subject, which is to be avoided from the irradiation, is included in the irradiation range of the radiation.

[0066] With the information processing apparatus **10** according to the embodiment of the disclosed technology, the irradiation range of the radiation at the position corresponding to the distance to the specific part of the subject in the irradiation axis direction is derived regardless of presence or absence of unevenness. Therefore, it is easy to determine whether or not the specific part is included in the irradiation range of the radiation. That is, with the information processing apparatus **10** according to the embodiment of the disclosed technology, it is possible to support the determination of whether or not the specific part is included in the irradiation range of the radiation.

[0067] In addition, as shown in FIGS. **4**A and **4**B, since the optical camera **30** is disposed at the position closer to the irradiation axis AX of the radiation with respect to the distance sensor **40**, a parallax between the optical image and the radiation image can be reduced. Therefore, it is easy to perform registration in a case where the information indicating the irradiation range of the radiation is displayed by superimposing the information on the optical image.

Second Embodiment

[0068] FIG. **11** is a functional block diagram showing an example of a functional configuration of the information processing apparatus **10** according to a second embodiment of the disclosed technology. The information processing apparatus **10** according to the second embodiment is different from the information processing apparatus **10** according to the above-described first embodiment in that a determination unit **17** is further provided.

[0069] The determination unit **17** determines whether or not the eyeball of the subject P, which is the specific part recognized by the specific part recognition unit **13**, is included in the irradiation range of the radiation derived by the irradiation range derivation unit **15**. In a case where the determination unit **17** determines that the eyeball of the subject P is included in the irradiation range of the radiation, the determination unit **17** issues an alert. The alert may be issued by, for example, display on the display **105**, output of a voice, light emission of a lamp, or the like.

[0070] FIG. **12** is a flowchart showing an example of a flow of processing performed by executing the processing program **110** by the CPU **101** of the information processing apparatus **10** according to the second embodiment. Since the processing from step S1 to step S6 is the same as the processing according to the above-described first embodiment, the description thereof will not be shown.

[0071] In step S7, the determination unit **17** determines whether or not the specific part (eyeball) recognized in step S3 is included in the irradiation range of the radiation derived in step S5. In a case where it is determined that the specific part (eyeball) is included in the irradiation range of the radiation, the processing proceeds to step S8, and in a case where it is determined that the specific part (eyeball) is not included in the irradiation range of the radiation, the present routine ends.

[0072] In step S8, the determination unit **17** issues an alert to notify that the specific part (eyeball) is included in the irradiation range of the radiation.

[0073] With the information processing apparatus **10** according to the second embodiment, in a case where the specific part (eyeball) of the subject P is included in the irradiation range of the radiation, an alert is issued, so that it is possible to suppress a risk that the specific part (eyeball) is irradiated with the radiation.

Third Embodiment

[0074] FIG. **13** is a functional block diagram showing an example of a functional configuration of the information processing apparatus **10** according to a third embodiment of the disclosed technology. The information processing apparatus **10** according to the third embodiment is different from the information processing apparatus **10** according to the above-described second embodiment in that an irradiation range controller **18** is further provided.

[0075] In a case where the eyeball of the subject P, which is the specific part recognized by the specific part recognition unit **13**, is included in the irradiation range of the radiation derived by the irradiation range derivation unit **15**, the irradiation range controller **18** controls the irradiation range of the radiation such that the specific part (eyeball) is not included in the irradiation range of the radiation in accordance with a command. The irradiation range controller **18** controls the irradiation range of the radiation by controlling an opening amount of an opening portion, through which the radiation passes, of the collimator **26** that forms the opening portion.

[0076] FIG. **14** is a perspective view showing an example of a configuration of the collimator **26**. The collimator **26** includes four blades **61**A, **61**B, **61**C, and **61**D. The blades **61**A to **61**D are plate-like members made of a material that shields radiation, such as lead or tungsten. Each blade is disposed such that one side surface of the blade **61**A and one side surface of the blade **61**B face each other and one side surface of the blade **61**C and one side surface of the blade **61**D face each other. The opening portion **60** having a rectangular shape in plan view is formed in a region surrounded by the blades **61**A to **61**D. The radiation emitted from the radiation tube **28** passes through the opening portion **60** to be emitted to the subject P.

[0077] Each of the blades **61**A to **61**D can be moved by a linear motion mechanism (not shown) including a motor, a rack, and a pinion. In a case where the radiation source unit **24** is in a posture shown in FIG. **2**, the blades **61**A and **61**B are movable in the Z direction, and the blades **61**C and **61**D are movable in the X direction. The irradiation range controller **18** controls an opening amount of the opening portion **60** by moving the blades **61**A to **61**D such that the specific part (eyeball) is not included in the irradiation range of the radiation. Movement amounts of the blades **61**A to **61**D for allowing the specific part (eyeball) to not be included in the irradiation range of the radiation can be derived from a positional relationship (distance) between the irradiation range of the radiation derived by the irradiation range derivation unit **15** and the specific part (eyeball) recognized by the specific part recognition unit **13**.

[0078] FIG. **15** is a flowchart showing an example of a flow of processing performed by executing the processing program **110** by the CPU **101** of the information processing apparatus **10** according to the third embodiment. Since the processing from step S1 to step S8 is the same as the processing according to the above-described second embodiment, the description thereof will not be shown.

[0079] In step S9, the irradiation range controller **18** determines presence or absence of a control command for controlling the irradiation range of the radiation. The control command of the irradiation range of the radiation is input to the information processing apparatus **10** by, for example, operating the input device **104** by a user. In a case where it is determined that there is the control command, the processing proceeds to step S10, and in a case where it is determined that there is no control command, the present routine ends.

[0080] In step S10, the irradiation range controller **18** controls the irradiation range of the radiation such that the specific part (eyeball) recognized in step S2 is not included in the irradiation range of the radiation. The irradiation range controller **18** controls the irradiation range of the radiation by controlling the opening amount of the opening portion **60** of the collimator **26**.

[0081] With the information processing apparatus **10** according to the third embodiment, in a case where a specific part of the subject P is included in an irradiation range of radiation, the irradiation range of the radiation is controlled such that the part is not included in the irradiation range of the radiation. Therefore, it is possible to reduce work burden related to the adjustment of the irradiation range for avoiding the irradiation of the radiation to the specific part.

[0082] In the above description, a case where the “specific part” is the eyeball has been described as an example, but the disclosed technology is not limited to thereto. For example, a part to be avoided from the irradiation of the radiation, such as a reproductive gland of the subject and a hand of the capturer, can be referred to as a “specific part”. The disclosed technology can also be applied in a case where a part other than the eyeball is referred to as a “specific part”. In addition, the disclosed technology is not limited to the videofluoroscopic examination of swallowing, and can be applied to any examination using a radiography apparatus.

[0083] In addition, in the above description, a case where the specific part recognition unit **13** recognizes the specific part from the optical image by using the discriminator constructed by machine learning has been described as an example. However, the specific part may be directly recognized by using a result learned from the optical image, or may be secondarily recognized in consideration of an anatomical geometrical relationship from a part such as a joint that is easily extracted directly.

[0084] In each of the above-described embodiments, for example, as a hardware structure of processing units that execute various types of processing, such as the optical image acquisition unit **11**, the distance image acquisition unit **12**, the specific part recognition unit **13**, the distance derivation unit **14**, the irradiation range derivation unit **15**, the display processing unit **16**, the determination unit **17**, and the irradiation range controller **18**, various processors shown below can be used. The above-described various processors include, for example, a programmable logic device (PLD) which is a processor having a changeable circuit configuration after manufacturing, such as an FPGA, and a dedicated electrical circuit which is a processor having a dedicated circuit configuration designed to execute specific processing, such as an application specific integrated

circuit (ASIC), in addition to the GPU and the CPU which is a general-purpose processor that executes software (programs) to function as various processing units, as described above.
[0085] One processing unit may be configured with one of the various types of processors, or a combination of two or more processors of the same type or different types (for example, a combination of a plurality of FPGAs, or a combination of a CPU and an FPGA). Further, a plurality of processing units may be configured with one processor.
[0086] As an example of configuring a plurality of processing units with one processor, first, there is a form in which, as typified by computers such as a client and a server, one processor is configured by combining one or more CPUs and software, and the processor functions as a plurality of processing units. A second example of the configuration is a form in which a processor that implements the functions of the entire system including a plurality of processing units using one integrated circuit (IC) chip is used. A representative example of this form is a system-on-chip (SoC). As described above, the various types of processing units are configured using one or more of the above-described various types of processors as a hardware structure.
[0087] Further, as the hardware structure of these various processors, more specifically, an electric circuit (circuitry), in which circuit elements such as semiconductor elements are combined, can be used.
[0088] Moreover, in the above-described embodiments described above, the aspect has been described in which the processing program **110** is stored (installed) in advance in the non-volatile memory **103**, but the disclosed technology is not limited to this. The processing program **110** may be provided in a form in which the programs are recorded in a recording medium such as a compact disc read only memory (CD-ROM), a digital versatile disc read-only memory (DVD-ROM), and a universal serial bus (USB) memory. Alternatively, a form may be employed in which the above-described processing program **110** is downloaded from an external device via the network.
[0089] Regarding the first to third embodiments, the following appendixes are further disclosed.
APPENDIX 1
[0090] An information processing apparatus comprising at least one processor, [0091] wherein the processor is configured to: [0092] recognize a specific part of a subject from an optical image obtained by capturing the subject; [0093] derive a distance to the recognized specific part using a distance image obtained by capturing the subject; and [0094] derive an irradiation range of radiation at a position corresponding to the derived distance in an irradiation axis direction.
APPENDIX 2
[0095] The information processing apparatus according to appendix 1, [0096] wherein the processor is configured to display information indicating the derived irradiation range on a display screen, by superimposing the information on the optical image.
APPENDIX 3
[0097] The information processing apparatus according to appendix 1 or 2, [0098] wherein the processor is configured to display the optical image and a radiation image side by side.
APPENDIX 4
[0099] The information processing apparatus according to any one of appendixes 1 to 3, [0100] wherein the processor is configured to issue an alert in a case in which the recognized specific part is included in the derived irradiation range.
APPENDIX 5
[0101] The information processing apparatus according to any one of appendixes 1 to 4, [0102] wherein the processor is configured to control the irradiation range of the radiation such that the recognized specific part is not included in the irradiation range of the radiation in accordance with a command, in a case where the recognized specific part is included in the derived irradiation range.
APPENDIX 6
[0103] The information processing apparatus according to appendix 5, [0104] wherein the processor is configured to control the irradiation range of the radiation by controlling an opening

amount of an opening portion, through which the radiation passes, of a collimator that forms the opening portion.
APPENDIX 7
[0105] The information processing apparatus according to any one of appendixes 1 to 6, [0106] wherein the specific part is an eyeball of the subject.
APPENDIX 8
[0107] A radiography system comprising: [0108] the information processing apparatus according to any one of appendixes 1 to 7; [0109] an optical camera that generates the optical image; [0110] a distance sensor that generates the distance image; and [0111] a radiation source unit that emits radiation.
APPENDIX 9
[0112] The radiography system according to appendix 8, [0113] wherein the optical camera is disposed outside an irradiation range of the radiation emitted from the radiation source unit.
APPENDIX 10
[0114] The radiography system according to appendix 8 or 9, [0115] wherein the optical camera is disposed at a position closer to an irradiation axis of the radiation emitted from the radiation source unit with respect to the distance sensor.
APPENDIX 11
[0116] The radiography system according to any one of appendixes 8 to 10, [0117] wherein the distance sensor is a ToF camera or a stereo camera.
APPENDIX 12
[0118] An information processing method executed by at least one processor included in an information processing apparatus, the method including a process comprising: [0119] recognizing a specific part of a subject from an optical image obtained by capturing the subject; [0120] deriving a distance to the recognized specific part using a distance image obtained by capturing the subject; and [0121] deriving an irradiation range of radiation at a position corresponding to the derived distance in an irradiation axis direction.
APPENDIX 13
[0122] A program that causes at least one processor included in an information processing apparatus, to execute a process comprising: [0123] recognizing a specific part of a subject from an optical image obtained by capturing the subject; [0124] deriving a distance to the recognized specific part using a distance image obtained by capturing the subject; and [0125] deriving an irradiation range of radiation at a position corresponding to the derived distance in an irradiation axis direction.

## Claims

**1**. An information processing apparatus comprising at least one processor, wherein the processor is configured to: recognize a specific part of a subject from an optical image obtained by capturing the subject; derive a distance to the recognized specific part using a distance image obtained by capturing the subject; and derive an irradiation range of radiation at a position corresponding to the derived distance in an irradiation axis direction.
**2**. The information processing apparatus according to claim 1, wherein the processor is configured to display information indicating the derived irradiation range on a display screen, by superimposing the information on the optical image.
**3**. The information processing apparatus according to claim 1, wherein the processor is configured to display the optical image and a radiation image side by side.
**4**. The information processing apparatus according to claim 1, wherein the processor is configured to issue an alert in a case where the recognized specific part is included in the derived irradiation range.

**5**. The information processing apparatus according to claim 1, wherein the processor is configured to control the irradiation range of the radiation such that the recognized specific part is not included in the irradiation range of the radiation in accordance with a command, in a case where the recognized specific part is included in the derived irradiation range.
**6**. The information processing apparatus according to claim 5, wherein the processor is configured to control the irradiation range of the radiation by controlling an opening amount of an opening portion, through which the radiation passes, of a collimator that forms the opening portion.
**7**. The information processing apparatus according to claim 1, wherein the specific part is an eyeball of the subject.
**8**. A radiography system comprising: the information processing apparatus according to claim 1; an optical camera that generates the optical image; a distance sensor that generates the distance image; and a radiation source unit that emits radiation.
**9**. The radiography system according to claim 8, wherein the optical camera is disposed outside an irradiation range of the radiation emitted from the radiation source unit.
**10**. The radiography system according to claim 8, wherein the optical camera is disposed at a position closer to an irradiation axis of the radiation emitted from the radiation source unit with respect to the distance sensor.
**11**. The radiography system according to claim 8, wherein the distance sensor is a ToF camera or a stereo camera.
**12**. An information processing method executed by at least one processor included in an information processing apparatus, the method including a process comprising: recognizing a specific part of a subject from an optical image obtained by capturing the subject; deriving a distance to the recognized specific part using a distance image obtained by capturing the subject; and deriving an irradiation range of radiation at a position corresponding to the derived distance in an irradiation axis direction.
**13**. A non-transitory computer-readable storage medium storing a program that causes at least one processor included in an information processing apparatus, to execute a process comprising: recognizing a specific part of a subject from an optical image obtained by capturing the subject; deriving a distance to the recognized specific part using a distance image obtained by capturing the subject; and deriving an irradiation range of radiation at a position corresponding to the derived distance in an irradiation axis direction.